

What is the effectiveness of PD-1 inhibitor monotherapy compared to combination therapy as first-line treatment for PD-L1-high metastatic non-small cell lung cancer in terms of overall survival and progression-free survival?

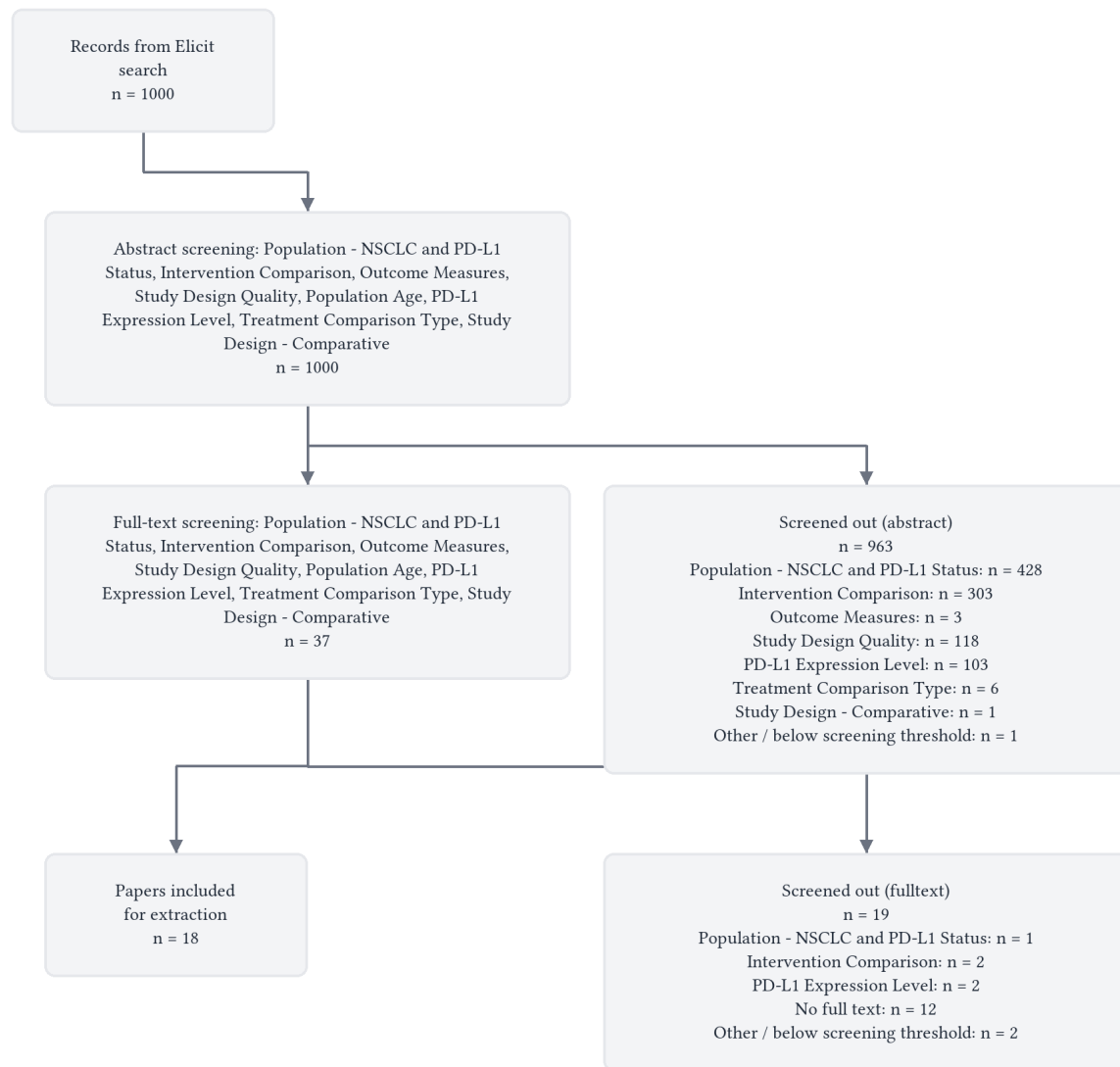
In PD-L1-high metastatic NSCLC, PD-1 inhibitor–chemotherapy combination therapy consistently improves progression-free survival over PD-1 inhibitor monotherapy, but overall survival is similar between the two strategies, with combination therapy carrying greater toxicity — though this conclusion rests entirely on indirect comparisons, as no direct head-to-head randomized trial has been completed.

Abstract

Across 18 included sources — predominantly network meta-analyses and indirect comparisons of randomized controlled trials — PD-1 inhibitor monotherapy and PD-1 inhibitor–chemotherapy combination therapy both significantly improve overall survival and progression-free survival relative to chemotherapy alone in patients with PD-L1-high (TPS $\geq 50\%$) metastatic NSCLC [1–3]. Combination therapy consistently demonstrates superior progression-free survival compared with monotherapy, with hazard ratios ranging from 0.55 to 0.81 across analyses [2, 4–7]. However, no statistically significant overall survival difference between the two strategies has been identified in the PD-L1-high subgroup: median OS is approximately 22–26 months with either approach [2, 4, 8, 9]. Combination therapy is associated with higher rates of grade 3–5 treatment-related adverse events [9–11].

No completed randomized trial has directly compared PD-1 inhibitor monotherapy with PD-1 inhibitor–chemotherapy in PD-L1-high disease; all comparisons are indirect, anchored through a chemotherapy common comparator [2, 4, 5]. The two available head-to-head RCTs evaluated other combination partners (ipilimumab and epacadostat) added to pembrolizumab and found no efficacy benefit [10, 12]. Consequently, whether the PFS advantage of chemo-immunotherapy translates into a meaningful long-term survival benefit in PD-L1-high patients, or whether monotherapy provides equivalent OS with lower toxicity, remains formally unresolved. Critical gaps include the absence of validated predictive biomarkers beyond PD-L1 TPS for differential treatment selection [9, 13], insufficient safety and quality-of-life data stratified by PD-L1 expression [6, 8], and underrepresentation of key clinical subgroups such as patients with liver metastases, brain metastases, or advanced age [1, 14].

Flow Diagram



Paper search

We performed a semantic search across over 138 million academic papers from the Elicit search engine, which includes all of Semantic Scholar and OpenAlex.

We ran this query: "What is the effectiveness of PD-1 inhibitor monotherapy compared to combination therapy as first-line treatment for PD-L1-high metastatic non-small cell lung cancer in terms of overall survival and progression-

free survival?”

The search returned 1,000 total results from Elicit.

We retrieved 1000 papers most relevant to the query for screening.

Screening

Abstract screening

We screened in sources based on their abstracts that met these criteria:

- **Population - NSCLC and PD-L1 Status:** Does this study include patients with histologically or cytologically confirmed metastatic (stage IV) non-small cell lung cancer with PD-L1-high expression ($\geq 50\%$ tumor proportion score or equivalent high expression threshold)?
- **Intervention Comparison:** Does this study compare PD-1 inhibitor monotherapy versus PD-1 inhibitor combination therapy as first-line treatment for metastatic disease?
- **Outcome Measures:** Does this study report overall survival and/or progression-free survival outcomes?
- **Study Design Quality:** Is this study a randomized controlled trial, systematic review, or meta-analysis (not a case report, case series, editorial, or conference abstract)?
- **Population Age:** Does this study include adult patients (≥ 18 years) rather than focusing on pediatric populations?
- **PD-L1 Expression Level:** Does this study focus on patients with PD-L1-high tumors rather than PD-L1-low or PD-L1-negative tumors?
- **Treatment Comparison Type:** Does this study compare different PD-1 inhibitor approaches (monotherapy vs combination) rather than comparing PD-1 inhibitors to non-PD-1 inhibitor treatments (e.g., chemotherapy alone, targeted therapy)?
- **Study Design - Comparative:** Does this study include comparison groups rather than being a single-arm study without comparators?

Papers that failed any strict criterion were automatically excluded. For the remaining papers, we considered all screening questions together and made a holistic judgement about whether to screen in each paper.

37 papers passed abstract screening and moved to full-text screening.

At abstract screening, the number of papers excluded for each primary reason was:

- **Population - NSCLC and PD-L1 Status:** n = 428
- **Intervention Comparison:** n = 303
- **Outcome Measures:** n = 3
- **Study Design Quality:** n = 118
- **PD-L1 Expression Level:** n = 103
- **Treatment Comparison Type:** n = 6
- **Study Design - Comparative:** n = 1
- **Other / below screening threshold:** n = 1

Full-text screening

We then screened papers based on their full text using these additional criteria:

- **Population - NSCLC and PD-L1 Status:** Does this study include patients with histologically or cytologically confirmed metastatic (stage IV) non-small cell lung cancer with PD-L1-high expression ($\geq 50\%$ tumor proportion score or equivalent high expression threshold)?
- **Intervention Comparison:** Does this study compare PD-1 inhibitor monotherapy versus PD-1 inhibitor combination therapy as first-line treatment for metastatic disease?
- **Outcome Measures:** Does this study report overall survival and/or progression-free survival outcomes?
- **Study Design Quality:** Is this study a randomized controlled trial, systematic review, or meta-analysis (not a case report, case series, editorial, or conference abstract)?
- **Population Age:** Does this study include adult patients (≥ 18 years) rather than focusing on pediatric populations?
- **PD-L1 Expression Level:** Does this study focus on patients with PD-L1-high tumors rather than PD-L1-low or PD-L1-negative tumors?
- **Treatment Comparison Type:** Does this study compare different PD-1 inhibitor approaches (monotherapy vs combination) rather than comparing PD-1 inhibitors to non-PD-1 inhibitor treatments (e.g., chemotherapy alone, targeted therapy)?
- **Study Design - Comparative:** Does this study include comparison groups rather than being a single-arm study without comparators?

Papers that failed any strict criterion were automatically excluded. For the remaining papers, we considered all screening questions together and made a holistic judgement about whether to include each paper in the final analysis.

18 papers passed full-text screening and moved to data extraction.

At full-text screening, the number of papers excluded for each primary reason was:

- **Population - NSCLC and PD-L1 Status:** n = 1
- **Intervention Comparison:** n = 2
- **PD-L1 Expression Level:** n = 2
- **No full text:** n = 12
- **Other / below screening threshold:** n = 2

Data extraction

We asked a large language model to extract each data column below from each paper. We gave the model the extraction instructions shown below for each column.

- **Treatment Comparison:**

Extract details about the specific PD-1 inhibitor treatments being compared for first-line therapy in PD-L1-high metastatic NSCLC, including:

- Specific PD-1 inhibitor monotherapy (drug name, dose, schedule)
- Specific combination therapy (PD-1 inhibitor + additional agents, doses, schedules)
- Whether this was a direct head-to-head comparison or indirect comparison
- Control/reference treatment if applicable

- **Study Population:**

Extract patient population characteristics relevant to PD-L1-high metastatic NSCLC first-line treatment, including:

- PD-L1 expression threshold used ($\geq 50\%$, $\geq 1\%$, etc.)
- Confirmation of metastatic/advanced stage NSCLC
- Confirmation of first-line treatment setting
- Histology (squamous, non-squamous, mixed)
- EGFR/ALK mutation status
- Sample size for each treatment arm

- **Overall Survival:**

Extract all overall survival data comparing PD-1 inhibitor monotherapy vs combination therapy in PD-L1-high metastatic NSCLC, including:

- Median OS for each treatment group (months)
- Hazard ratio with 95% confidence intervals
- P-value for OS comparison
- Follow-up duration
- Number of events/deaths in each group
- Any subgroup OS analyses by PD-L1 expression level

- **Progression-Free Survival:**

Extract all progression-free survival data comparing PD-1 inhibitor monotherapy vs combination therapy in PD-L1-high metastatic NSCLC, including:

- Median PFS for each treatment group (months)
- Hazard ratio with 95% confidence intervals
- P-value for PFS comparison
- PFS assessment method (RECIST, investigator vs independent review)
- Any subgroup PFS analyses by PD-L1 expression level

- **Study Design:**

Extract study methodology details that affect interpretation of monotherapy vs combination therapy comparisons, including:

- Study type (RCT, meta-analysis, network meta-analysis, observational)
- Randomization and blinding approach
- Primary vs secondary endpoint analysis
- Statistical methods used for comparison
- Risk of bias assessment if provided

- **Safety Profile:**

Extract safety data comparing PD-1 inhibitor monotherapy vs combination therapy in PD-L1-high metastatic NSCLC, including:

- Grade 3-5 adverse events (percentage in each group)
- Treatment-related deaths (percentage in each group)
- Treatment discontinuation rates due to toxicity
- Specific immune-related adverse events if reported
- Any safety-related study modifications or early stopping

- **Patient Characteristics:**

Extract baseline patient characteristics that may moderate effectiveness of monotherapy vs combination therapy in PD-L1-high metastatic NSCLC, including:

- Age distribution and median age
- Performance status (ECOG 0 vs 1+)
- Smoking history
- Presence of brain metastases
- Presence of liver metastases
- Tumor burden indicators if reported

- **Key Findings:**

Extract the study's main conclusions about effectiveness of PD-1 inhibitor monotherapy vs combination therapy for PD-L1-high metastatic NSCLC, including:

- Authors' primary conclusion about comparative effectiveness
- Clinical significance interpretation by authors
- Recommendations for clinical practice
- Limitations or caveats mentioned by authors
- Future research recommendations if provided

Summary of Current Evidence

The included studies consistently demonstrate that both PD-1 inhibitor monotherapy and PD-1 inhibitor-based combination therapy (typically with platinum-based chemotherapy) significantly improve overall survival and progression-free survival compared with chemotherapy alone in patients with PD-L1-high (TPS $\geq 50\%$) metastatic NSCLC [1–3]. A recurring finding across network meta-analyses and indirect comparisons is that combination therapy yields superior objective response rates and progression-free survival relative to monotherapy in PD-L1-high patients [4–7], whereas overall survival differences between these two strategies are not statistically significant in this subgroup [2, 4, 8, 15]. The KEYNOTE-598 trial established that adding ipilimumab to pembrolizumab confers no efficacy benefit and increases toxicity [10, 11]. Current guidelines accordingly treat pembrolizumab monotherapy and pembrolizumab–chemotherapy as complementary first-line options for PD-L1-high disease, with selection guided by clinical context [9].

Evidence Map

Study	Study Type	Research Focus	Key Finding	Identified Limitation
R. Herbst et al., 2021	Network meta-analysis of RCTs [1]	Comparison of multiple CIT regimens vs chemotherapy in PD-L1-high NSCLC across histology networks [1]	Atezolizumab, pembrolizumab, and cemiplimab monotherapies significantly improved OS vs chemotherapy in the mixed-histology network; pembrolizumab plus chemotherapy improved OS in non-squamous network [1]	Lack of individual patient data for most trials; heterogeneity in PD-L1 assay methods; most included trials were open-label [1, 1]
Lingling Li et al., 2020	Systematic review with indirect comparison [16]	IO monotherapy vs IO plus chemotherapy as first-line treatment for advanced NSCLC [16]	IC yielded significantly better PFS than IO when PD-L1 $\geq 50\%$ (HR 1.81, 95% CI 1.18–2.78); no significant OS difference in PD-L1 $\geq 50\%$ subgroup [16, 16]	Indirect comparison only; no head-to-head trial data; limited ability to assess treatment-related deaths across strategies [16]
Patryk Bachurski et al., 2026	Narrative review of RCTs, NMAs, and real-world cohorts [9]	Pembrolizumab monotherapy vs pembrolizumab–chemotherapy in driver-negative metastatic NSCLC [9]	Combination therapy improves PFS (HR 0.59, 95% CI 0.43–0.81) but does not clearly improve OS compared with monotherapy in PD-L1-high disease [9, 9]	Reliance on cross-trial and real-world comparisons; simple PD-L1 cutoffs insufficient for full individualization [9]

Study	Study Type	Research Focus	Key Finding	Identified Limitation
J. H. Kim et al., 2021	Bayesian network meta-analysis [2]	Frontline treatments for advanced NSCLC with high PD-L1 expression [2]	ICI plus chemotherapy failed to demonstrate significant OS superiority over ICI monotherapy (HR 0.978, 95% CrI 0.771–1.259) [2]; ICI plus chemotherapy significantly improved PFS vs monotherapy (HR 0.571, 95% CrI 0.454–0.709) [2]	Aggregated rather than individual patient data; limited studies in certain treatment nodes; variability in PD-L1 measurement methods [2]
Shibo Huang et al., 2024	Systematic review and network meta-analysis [4]	Pembrolizumab plus chemotherapy vs pembrolizumab monotherapy for metastatic NSCLC [4]	In PD-L1 $\geq 50\%$, combination improved PFS (median 10.41 vs 7.41 months, HR 0.81, P=0.02) but not OS (median 22.54 vs 22.62 months, HR 0.89, P=0.24) [4, 4]	Only five studies included; potential errors in reconstructing individual patient data from Kaplan-Meier curves; heterogeneity in baseline characteristics [4]
Yixin Zhou et al., 2019	Meta-analysis with indirect comparison [5]	Pembrolizumab plus chemotherapy vs pembrolizumab monotherapy in PD-L1 TPS $\geq 50\%$ advanced NSCLC [5]	Combination therapy superior for ORR (RR 1.62) and PFS (HR 0.55, 95% CI 0.32–0.97); trend toward improved OS but not significant (HR 0.76, P=0.184) [5, 5]	Indirect comparison relying on published aggregate data; short median follow-up (7.8–23.9 months); lack of head-to-head trial data [5]
Hengrui Liang et al., 2020	Bayesian network meta-analysis [3]	Efficacy and safety of PD-1/L1 inhibitors in first-line advanced NSCLC [3]	Pembrolizumab plus chemotherapy (HR 0.39) and pembrolizumab monotherapy (HR 0.67) both significantly improved PFS vs chemotherapy in PD-L1 $\geq 50\%$ [3]	Indirect comparisons only; subgroup analyses based solely on PD-L1 expression may lack accuracy; focus limited to high-grade adverse events [3]

Study	Study Type	Research Focus	Key Finding	Identified Limitation
Xiang Li et al., 2021	Network meta-analysis of RCTs [8]	PD-1/PD-L1 inhibitors plus chemotherapy vs monotherapy in advanced NSCLC [8]	PD-1 inhibitors plus chemotherapy improved PFS (HR 0.72, 95% CI 0.63–0.83) but not OS (HR 0.94, 95% CI 0.82–1.07) in PD-L1 $\geq 50\%$ [8, 8]	Distinction between PD-1 and PD-L1 inhibitor classes complicates pooled interpretation; ongoing trials may alter conclusions; safety data not stratified by PD-L1 level [8]
M. Boyer et al., 2021	Phase III RCT (KEYNOTE-598) [10]	Pembrolizumab plus ipilimumab vs pembrolizumab monotherapy in PD-L1 TPS $\geq 50\%$ metastatic NSCLC [10]	No improvement in OS (HR 1.08, P=0.74) or PFS (HR 1.06, P=0.72) with addition of ipilimumab; greater toxicity with combination (grade 3–5 AEs: 62.4% vs 50.2%) [10, 10, 10]	Evaluates pembrolizumab–ipilimumab rather than pembrolizumab–chemotherapy; study stopped early for futility; does not assess pembrolizumab–chemotherapy combinations [10]
M. Siciliano et al., 2022	Systematic review and Bayesian NMA [14]	Ranking of first-line ICI-based regimens for non-oncogene-addicted NSCLC [14]	Cemiplimab ranked first for OS in PD-L1 $>50\%$ (HR 0.33, 95% CrI 0.14–0.75); atezolizumab/bevacizumab ranked first for PFS [14, 14]	Heterogeneity in included data; potential biases from crossover in RCTs; limited safety profile characterization in PD-L1-high subgroup [14]
Zheng Chuangjie et al., 2022	Systematic review and network meta-analysis [17]	ICI alone vs ICI plus chemotherapy in PD-L1 $\geq 50\%$ NSCLC [17]	Primary outcome was PFS; OS data not reported [17, 17]	No sensitivity analysis performed; limited detail on included studies and outcomes; OS not evaluated [17]

Study	Study Type	Research Focus	Key Finding	Identified Limitation
T. Peng et al., 2021	Network meta-analysis [13]	First-line ICIs for advanced NSCLC across PD-L1 subgroups [13]	Pembrolizumab plus platinum-based chemotherapy ranked best for PFS in PD-L1 $\geq 50\%$ (HR 0.39, 95% CI 0.27–0.56); atezolizumab ranked best for OS in this subgroup [13, 13]	Heterogeneity from different treatment regimens; no data on tumor mutational burden; limited safety reporting [13]
R. Pathak et al., 2020	Network meta-analysis [6]	Chemoimmunotherapy vs ICI for first-line advanced NSCLC [6]	Chemo-ICI showed significantly improved PFS (HR 0.59, 95% CrI 0.41–0.85) and ORR vs single-agent ICI in PD-L1-high tumors; no significant OS difference [6, 6]	No safety data assessed; potential misclassification bias from differences in PD-L1 testing assays; no assessment for publication bias [6]
T. Tokito et al., 2024	Phase II RCT [12]	Epacadostat plus pembrolizumab vs pembrolizumab alone in PD-L1 $\geq 50\%$ metastatic NSCLC [12]	No improvement in ORR (32.5% vs 39.0%), PFS (6.7 vs 6.2 months, HR 1.10), or OS (HR 0.74, P=0.205) with epacadostat addition [12, 12]	Small sample size (n=154); short follow-up (median ~7 months); study downsized from phase III to phase II and discontinued early [12]
Yimin Wang et al., 2022	Network meta-analysis [7]	ICI alone vs chemo-ICI in high PD-L1 expression NSCLC [7]	Chemo-ICI significantly improved ORR (OR 1.7) and PFS (HR 0.59, 95% CI 0.48–0.74) vs ICI; no significant OS difference (HR 0.82, 95% CI 0.6–1.1) [7, 7]	Reliance on published aggregate results rather than individual patient data; limited subgroup data for squamous histology and PD-L1 inhibitor subsets [7]

Study	Study Type	Research Focus	Key Finding	Identified Limitation
Rui Cao et al., 2019	Meta-analysis [15]	ICIs vs chemotherapy vs combination as first-line treatment for advanced NSCLC [15]	In PD-L1 $\geq 50\%$, similar OS between pembrolizumab monotherapy and combination therapy; combination improved PFS and ORR [15, 15]	Study-level evidence with potential inter-study heterogeneity; no detailed safety data for PD-L1-high subgroup [15]
D. Rodríguez Abreu et al., 2022	Phase III RCT follow-up (KEYNOTE-598) [11]	3-year follow-up of pembrolizumab plus ipilimumab vs pembrolizumab plus placebo in PD-L1 TPS $\geq 50\%$ [11]	No OS or PFS benefit from adding ipilimumab; median PFS 8.2 vs 8.4 months (HR 0.99); higher grade 3–5 treatment-related AEs with combination (35.1% vs 20.3%) [11, 11]	Ipilimumab and placebo discontinued per DMC recommendation; subsequent pembrolizumab monotherapy in both arms complicates long-term analysis; does not evaluate chemo-immunotherapy combinations [11]
DataSheet1.docx (cost-effectiveness analysis)	Markov model with network meta-analysis [18]	Cost-effectiveness of cemiplimab, pembrolizumab, and atezolizumab monotherapies and pembrolizumab–chemotherapy in PD-L1 $\geq 50\%$ NSCLC [18]	Cemiplimab was cost-effective compared with pembrolizumab and dominant over atezolizumab; pembrolizumab plus chemotherapy had ICERs of \$190,994–\$393,359/QALY vs monotherapies [18]	No head-to-head trial data; relied on modeled estimates from NMA; histology not specified; limited patient characteristic granularity [18]

The evidence base is dominated by network meta-analyses and indirect comparisons (14 of 18 sources), with only three primary RCTs (KEYNOTE-598 and its follow-up, and the epacadostat trial) providing direct head-to-head data, none of which compared pembrolizumab monotherapy with pembrolizumab–chemotherapy [10–12]. The NMAs consistently converge on the finding that combination therapy improves PFS (hazard ratios ranging from 0.55 to 0.81 across analyses) but does not significantly improve OS in PD-L1-high disease [2, 4, 6, 8]. However, almost all comparisons between monotherapy and chemo-immunotherapy are indirect, linked through a common chemotherapy comparator, which introduces structural uncertainty that cannot be resolved without a direct randomized comparison.

Gaps in the Evidence Base

The most consequential gap is the absence of any completed, adequately powered, randomized controlled trial directly comparing PD-1 inhibitor monotherapy with PD-1 inhibitor–chemotherapy in patients with PD-L1 TPS $\geq 50\%$. All efficacy comparisons between these strategies in the included literature derive from indirect methods — network meta-analyses anchored to a chemotherapy common comparator [2, 4, 5, 7] — or from cross-trial real-world comparisons subject to substantial confounding [9]. The only direct head-to-head RCTs examined pembrolizumab–ipilimumab versus pembrolizumab alone (KEYNOTE-598) [10] and pembrolizumab–epacadostat versus pembrolizumab alone [12], neither of which evaluated the clinically relevant pembrolizumab–chemotherapy comparator. This means that the central clinical question — whether the PFS advantage of combination therapy translates into meaningful long-term survival benefit or merely shifts disease control forward without altering the OS trajectory — remains formally unanswered. The consequence is that clinician decision-making between monotherapy and chemo-immunotherapy in PD-L1-high disease continues to rely on indirect evidence and clinical judgment rather than definitive trial data.

Second, the evidence base lacks adequate characterization of patient subgroups most likely to benefit differentially from monotherapy versus combination therapy. While several sources note that tumor burden, performance status, age, and histology may modify the relative benefit of these strategies [5, 9, 10], no included study provides prospective, biomarker-stratified subgroup analyses of monotherapy versus chemo-immunotherapy within the PD-L1-high population. Most NMAs do not report baseline characteristics such as liver metastases, brain metastases, or tumor burden in sufficient detail to perform meaningful moderator analyses [2, 6, 8, 13]. Only two studies explicitly discuss liver metastases as a clinical modifier [1, 14], and brain metastases subgroups are reported in only a few analyses [10, 12, 14]. The absence of validated predictive biomarkers beyond PD-L1 TPS — such as tumor mutational burden, gene expression profiles, or composite clinical-molecular scores — limits the ability to identify patients for whom the added toxicity of chemotherapy is justified by a genuine survival benefit versus those for whom monotherapy provides equivalent long-term outcomes with better tolerability [9, 13].

Third, safety data comparing monotherapy and combination therapy in the PD-L1-high subgroup specifically are sparse. Most NMAs either do not report adverse events stratified by PD-L1 expression [6–8] or provide only aggregate safety data across PD-L1 strata [3, 13]. Detailed reporting of treatment-related deaths, discontinuation rates, and immune-related adverse event profiles is available from only a few studies [10–12], and these evaluate combination partners (ipilimumab, epacadostat) other than chemotherapy. The consequence is that the incremental toxicity burden of adding chemotherapy to pembrolizumab — particularly in older patients and those with ECOG performance status 1 — cannot be precisely quantified for the PD-L1-high population from existing data, weakening the risk-benefit assessment that underpins treatment selection. Long-term quality-of-life and patient-reported outcome data are similarly absent from all included sources, leaving the tolerability comparison between strategies incomplete.

Recommendations for Future Research

- **Prospective randomized non-inferiority trial of pembrolizumab monotherapy versus pembrolizumab–chemotherapy in PD-L1 TPS $\geq 50\%$ NSCLC.** The trial should enroll treatment-naïve patients with metastatic NSCLC, PD-L1 TPS $\geq 50\%$, and no targetable driver alterations. The primary endpoint should be overall survival with a pre-specified non-inferiority margin (e.g., HR upper bound 1.20). Stratification by tumor burden, histology, and performance status would enable prospective subgroup analyses. This design is justified by the consistent NMA signal of equivalent OS between strategies [2, 4, 8] and addresses the most critical evidence gap. The EA5163/S1709 INSIGNA trial has explored related sequencing questions [10], but a definitive non-inferiority comparison remains needed.

- **Biomarker-stratified observational cohort study integrating tumor mutational burden, gene expression signatures, and composite clinical scores.** Given the repeated observation that PD-L1 TPS alone is insufficient for treatment individualization [9, 13], a prospective registry enrolling patients receiving either monotherapy or chemo-immunotherapy in routine practice should collect baseline tumor mutational burden, inflammatory gene signatures, neutrophil-to-lymphocyte ratio, and imaging-based tumor burden metrics. The primary objective would be to develop and validate a predictive model for differential OS benefit. This observational design is feasible because both strategies are already in widespread clinical use, and the key barrier is systematic biomarker collection rather than treatment randomization.
- **Systematic safety comparison with patient-reported outcomes in PD-L1-high patients receiving monotherapy versus chemo-immunotherapy.** A multi-center prospective cohort incorporating standardized adverse event grading (CTCAE v5), immune-related adverse event phenotyping, treatment discontinuation tracking, and validated patient-reported outcome instruments (e.g., EORTC QLQ-LC13) would address the current absence of quality-of-life data [9, 10]. Focus should include older adults (≥ 75 years) and patients with ECOG PS 1–2, populations frequently underrepresented in pivotal trials but commonly encountered in practice [9].
- **Subgroup-focused analyses of existing trial data for patients with liver metastases, brain metastases, and high tumor burden.** Pooled individual patient data meta-analysis of KEYNOTE-024, KEYNOTE-042, KEYNOTE-189, and KEYNOTE-407 could provide sufficiently powered subgroup comparisons that individual NMAs cannot achieve with aggregate data [1, 14]. Given that liver metastases are associated with poorer immunotherapy outcomes and brain metastases represent a growing clinical subpopulation, the absence of stratified efficacy data for these groups directly limits evidence-based treatment selection.
- **Cost-effectiveness analysis incorporating real-world effectiveness data and quality-of-life utilities.** The single included cost-effectiveness study relied on modeled clinical inputs from NMA rather than observed real-world outcomes [18]. A pragmatic economic evaluation using real-world data from health systems in diverse geographic settings — including non-US populations underrepresented in the current evidence base — would improve the generalizability of value assessments, particularly given the high incremental cost-effectiveness ratios reported for pembrolizumab–chemotherapy versus monotherapy ICIs [18].

References

1. Herbst R, Jassem J, Abogunrin S, et al (2021) A Network Meta-Analysis of Cancer Immunotherapies Versus Chemotherapy for First-Line Treatment of Patients With Non-Small Cell Lung Cancer and High Programmed Death-Ligand 1 Expression. *Frontiers in Oncology*. <https://doi.org/10.3389/fonc.2021.676732>
2. Kim JH, Jeong S, Lee J-J, et al (2021) A Bayesian Network Meta-Analysis of First-Line Treatments for Non-Small Cell Lung Cancer with High Programmed Death Ligand-1 Expression. *Journal of Clinical Medicine*. <https://doi.org/10.3390/jcm11061492>
3. Liang H, Lin G, Wang W, et al (2020) Feasibility and safety of PD-1/L1 inhibitors for non-small cell lung cancer in front-line treatment: a Bayesian network meta-analysis. *Translational Lung Cancer Research*. <https://doi.org/10.21037/tlcr.2020.02.14>

4. Huang S, Huang Z, Huang X, et al (2024) Comparative long-term outcomes of pembrolizumab plus chemotherapy versus pembrolizumab monotherapy as first-line therapy for metastatic non-small-cell lung cancer: a systematic review and network meta-analysis. *Frontiers in Immunology*. <https://doi.org/10.3389/fimmu.2024.1375136>
5. Zhou Y, Lin Z, Zhang X, et al (2019) First-line treatment for patients with advanced non-small cell lung carcinoma and high PD-L1 expression: pembrolizumab or pembrolizumab plus chemotherapy. *Journal of Immunotherapy for Cancer*. <https://doi.org/10.1186/s40425-019-0600-6>
6. Pathak R, Lopes G de L, Yu H, et al (2020) Comparative efficacy of chemoimmunotherapy versus immunotherapy for advanced non-small cell lung cancer: A network meta-analysis of randomized trials. *Cancer*. <https://doi.org/10.1002/cncr.33269>
7. Wang Y, Han H, Zhang F, et al (2022) Immune checkpoint inhibitors alone vs immune checkpoint inhibitors—combined chemotherapy for NSCLC patients with high PD-L1 expression: a network meta-analysis. *British Journal of Cancer*. <https://doi.org/10.1038/s41416-022-01832-4>
8. Li X, Yan S, Yang J, et al (2021) Efficacy and Safety of PD-1/PD-L1 Inhibitors Plus Chemotherapy Versus PD-1/PD-L1 Inhibitors in Advanced Non-Small Cell Lung Cancer: A Network Analysis of Randomized Controlled Trials. *Frontiers in Oncology*. <https://doi.org/10.3389/fonc.2020.574752>
9. Bachurski P, Chmiel G, Kalinowski P, et al (2026) Optimal First-Line Use of Pembrolizumab in Metastatic Driver-Negative NSCLC: Monotherapy Versus Chemo-Immunotherapy (Evidence 2022–2025). *Quality in Sport*. <https://doi.org/10.12775/qs.2026.51.68252>
10. Boyer M, Şendur M, Rodríguez-Abreu D, et al (2021) Pembrolizumab Plus Ipilimumab or Placebo for Metastatic Non-Small-Cell Lung Cancer With PD-L1 Tumor Proportion Score $\geq 50\%$: Randomized, Double-Blind Phase III KEYNOTE-598 Study. *Journal of Clinical Oncology*. <https://doi.org/10.1200/JCO.20.03579>
11. Abreu DR, Reck M, Sendur N, et al (2022) 6MO Pembrolizumab plus ipilimumab or placebo in previously untreated metastatic NSCLC with PD-L1 tumor proportion score $\geq 50\%$: KEYNOTE-598 3-year follow-up. *Annals of Oncology*. <https://doi.org/10.1016/j.annonc.2022.02.015>
12. Tokito T, Kolesnik O, Sørensen J, et al (2024) Epacadostat plus pembrolizumab versus placebo plus pembrolizumab as first-line treatment for metastatic non-small cell lung cancer with high levels of programmed death-ligand 1: a randomized, double-blind phase 2 study. *BMC Cancer*. <https://doi.org/10.1186/s12885-023-11203-8>
13. Peng T, Lin H-H, Tsai F-P, Wu T (2021) Immune checkpoint inhibitors for first-line treatment of advanced non-small-cell lung cancer: A systematic review and network meta-analysis. *Thoracic Cancer*. <https://doi.org/10.1111/1759-7714.14148>
14. Siciliano M, Caridà G, Ciliberto D, et al (2022) Efficacy and safety of first-line checkpoint inhibitors-based treatments for non-oncogene-addicted non-small-cell lung cancer: a systematic review and meta-analysis. *ESMO Open*. <https://doi.org/10.1016/j.esmoop.2022.100465>

15. Cao R, Ma J, Zhang S, et al (2019) Rational application of the first-line chemotherapy and immune checkpoint inhibitors in advanced nonsmall cell lung cancer: A meta-analysis. *Cancer Medicine*. <https://doi.org/10.1002/cam4.2407>
16. Li L, Xu F, Chen Y, et al (2020) Indirect comparison between immunotherapy alone and immunotherapy plus chemotherapy as first-line treatment for advanced non-small cell lung cancer: a systematic review. *BMJ Open*. <https://doi.org/10.1136/bmjopen-2019-034010>
17. chuangjie zheng, jiahao tao, linheng jiang, et al (2022) Immune checkpoint inhibitors, alone or in combination with chemotherapy, as first-line treatment for advanced non-small cell lung cancer in patients with PD-L1 $\geq$ 50%: a systematic review and network meta-analysis. <https://doi.org/10.37766/inplasy2022.4.0011>
18. DataSheet1.docx. <https://doi.org/10.3389/fphar.2021.788569.s001>